Name of the Student: _____ **Roll no:** _____

CLASS: - T.E (CSE) **Division:** **Course: - PCC305CSE- Machine Learning Lab**

Experiment No. 07

**** Support Vector Machine Classifiers****

Marks: /10

Date of Performance: / /2026 **Sign with Date:**

Aim: Support Vector Machine Classifiers with Linear and Nonlinear Kernels.

Lab Objective: To build, optimize, and evaluate Support Vector Machine (SVM) models using both Linear and Nonlinear (Radial Basis Function - RBF, Polynomial) kernels on an open-access benchmark dataset, and to contrast their classification boundaries and diagnostic performance using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Problem Statement:

Utilize Support Vector Machine classifiers with linear and nonlinear kernels on a suitable open-access dataset and compare classification performance using appropriate evaluation metrics.

1. Theory: The Optimal Hyperplane & Margin

Support Vector Machines identify the decision boundary (hyperplane) that maximizes the geometric margin between two classes. The margin is given by:

$$\text{Margin} = 2/\|w\|$$

Maximizing the margin corresponds to minimizing $_{21}\|w\|_2$, subject to the classification constraint for all samples:

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$$

Where ξ_i are slack variables and C is the regularization parameter balancing margin width against misclassification rate.

2. The Kernel Trick

When data points are not linearly separable in the input space, the **Kernel Trick** projects them into a higher-dimensional feature space $\phi(x)$ without explicitly computing coordinates in that high-dimensional space:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

- **Linear Kernel:** Suitable for linearly separable data or high-dimensional text data.

  $$K(x_i, x_j) = x_i^T x_j$$

- **Radial Basis Function (RBF) Kernel:** Creates localized, bell-shaped non-linear decision boundaries.

  $$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|_2)$$

  - Small γ: High variance, smooth wide boundaries.
  - Large γ: Low variance, tight complex boundaries around individual support vectors.

- **Polynomial Kernel:** Models feature interactions up to degree d.

  $$K(x_i, x_j) = (\gamma x_i^T x_j + r)^d$$

3. Dataset Selection

- **Dataset: Heart Disease Dataset** (UCI ML Repository / Scikit-Learn built-in).
- **Target:** Binary classification (Malignant vs. Benign).
- **Features:** 30 continuous dimensional descriptors of cell nuclei.

4. Algorithm & Execution Workflow

INPUT : Feature Matrix X, Label Vector y, Kernels List = ['linear', 'rbf', 'poly']
OUTPUT : Trained SVM models, Support Vector counts, Decision Boundary Plots, Performance Metrics

STEP 1: Load the open-access dataset and split 80% train / 20% test using stratified sampling.
STEP 2: Standardize features using StandardScaler (crucial for distance-based margins).
STEP 3: For each kernel in Kernels List:
    3.1 Initialize SVC(kernel=k, C=1.0).
    3.2 Fit model on scaled training subset.
    3.3 Record the number and coordinates of support vectors.
    3.4 Predict classes and probabilities on the unseen test subset.
    3.5 Compute Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
STEP 4: Tabulate comparative metric scores across all kernels.
STEP 5: Visualize 2D decision surfaces to contrast linear vs non-linear boundaries.

Questions:

1. What are Support Vectors, and what role do non-support vectors play in determining the decision boundary?

2. Why is feature scaling non-negotiable for Support Vector Classifiers?

3. How does the hyperparameter C control the bias–variance trade-off in SVM?

4. What does the γ (gamma) parameter represent in the RBF kernel?

5. When should one choose a Linear kernel over an RBF kernel?

Attendance and Timely Submission (10)	Execution (10)	Observations and Records (10)	Journal Documentation (10)	Oral (10)

Signature of Subject Teacher